Report-3

Based on black-box testing against the running QuickCart server, the following 26 explicit bugs were observed deviating from the QuickCart specification.

Bug 1: Profile Phone Number Accepts Alphabetical Characters

- **Endpoint tested:** `PUT /api/v1/profile`
- **Request payload:**
  - **Method:** PUT
  - **URL:** `http://localhost:8080/api/v1/profile`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** `{"name": "Alice", "phone": "abcdefghij"}`
- **Expected result:** 400 Bad Request (phone must be exactly 10 digits)
- **Actual result observed:** 200 OK
- **Failure detail:** Non-numeric phone input was accepted and profile update succeeded.

Bug 2: Valid Address Addition Rejected Improperly

- **Endpoint tested:** `POST /api/v1/addresses`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/addresses`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** `{"label": "HOME", "street": "123 Main St", "city": "Metropolis", "pincode": "123456"}`
- **Expected result:** 200 OK (valid address should be created)
- **Actual result observed:** 400 Bad Request
- **Failure detail:** Valid input matching documented constraints is rejected.

Bug 3: Street Minimum Bound Calculation Error (5 Characters Rejected)

- **Endpoint tested:** `POST /api/v1/addresses`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/addresses`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`

  - **Body:** `{"label": "HOME", "street": "12345", "city": "Metropolis", "pincode": "123456"}`

- **Expected result:** 200 OK (street length 5 is lower inclusive bound)
- **Actual result observed:** 400 Bad Request
- **Failure detail:** Lower boundary value for street length is rejected.

Bug 4: Street Maximum Bound Calculation Error (100 Characters Rejected)

- **Endpoint tested:** `POST /api/v1/addresses`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/addresses`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** `{"label": "HOME", "street": "AAAA

- **Actual result observed:** 200 OK
- **Failure detail:** Pincode longer than 6 digits is accepted.

Bug 7: Cart Add with Exact Zero Quantity Accepted

- **Endpoint tested:** `POST /api/v1/cart/add`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/cart/add`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** $\{\texttt{"product}_id\text{"} : 1, \text{"}quantity\text{"} : 0\}$
- **Expected result:** 400 Bad Request (quantity must be at least 1)
- **Actual result observed:** 200 OK
- **Failure detail:** Quantity 0 is accepted in cart add.

Bug 8: Cart Add with Negative Quantity Accepted

- **Endpoint tested:** `POST /api/v1/cart/add`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/cart/add`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** $\{\texttt{"product}_id\text{"} : 1, \text{"}quantity\text{"} : -5\}$
- **Expected result:** 400 Bad Request (quantity must be at least 1)
- **Actual result observed:** 200 OK
- **Failure detail:** Negative quantity is accepted in cart add.

Bug 9: Updating Cart that Doesn't Exist Returns Incorrect Status

- **Endpoint tested:** `POST /api/v1/cart/update`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/cart/update`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** $\{\texttt{"product}_id\text{"} : 999999, \text{"}quantity\text{"} : 2\}$
- **Expected result:** 404 Not Found (updating non-existent cart item should fail)
- **Actual result observed:** 200 OK
- **Failure detail:** Update call succeeds for product not present in cart.

Bug 10: Applying Invalid Coupon Code Returns 400 instead of 404

- **Endpoint tested:** `POST /api/v1/coupon/apply`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/coupon/apply`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** `{"code": "INVALID`$_{C}ODE_{1}23$"}
- **Expected result:** 404 Not Found (invalid/non-existent coupon should be treated as missing resource)
- **Actual result observed:** 400 Bad Request
- **Failure detail:** Server returns 400 instead of not-found semantics.

Bug 11: Review Created with Zero Rating

- **Endpoint tested:** `POST /api/v1/products/1/reviews`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/products/1/reviews`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** `{"rating": 0, "comment": "Okayish"}`
- **Expected result:** 400 Bad Request (rating must be in range 1-5)
- **Actual result observed:** 200 OK
- **Failure detail:** Out-of-range low rating is accepted.

Bug 12: Review Created with Rating Over Boundaries Limitation (6)

- **Endpoint tested:** `POST /api/v1/products/1/reviews`
- **Request payload:**
  - **Method:** POST
  - **URL:** `http://localhost:8080/api/v1/products/1/reviews`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
  - **Body:** `{"rating": 6, "comment": "Okayish"}`
- **Expected result:** 400 Bad Request (rating must be in range 1-5)
- **Actual result observed:** 200 OK
- **Failure detail:** Out-of-range high rating is accepted.

Bug 13: Creating a Review for a Non-Existent Product Appears Successful

- **Endpoint tested:** `POST /api/v1/products/{product`$_{i}d\}/reviews$

- `Request payload:`

  - **Method:** POST
  - **URL:** http://localhost:8080/api/v1/products/999999/reviews
  - **Headers:** {"X-Roll-Number": "123", "X-User-ID": "1"}
  - **Body:** {"rating": 4, "comment": "Nice"}
- **Expected result:** 404 Not Found (cannot create review for missing product)
- **Actual result observed:** 200 OK
- **Failure detail:** Review creation succeeds even for non-existent product id.

Bug 14: Product Price Mismatch Between User and Admin Endpoints

- **Endpoints tested:** GET /api/v1/products, GET /api/v1/admin/products
- **Request payload:**
  - **Method:** GET
  - **URL 1:** http://localhost:8080/api/v1/products
  - **URL 2:** http://localhost:8080/api/v1/admin/products
  - **Headers URL 1:** {"X-Roll-Number": "123", "X-User-ID": "1"}
  - **Headers URL 2:** {"X-Roll-Number": "123"}
- **Expected result:** For the same product$_id, the$price$should be identical on both endpoints.$
- **Actual result observed:** Price mismatch detected for same product (100 from user endpoint vs 95 from admin endpoint).

Bug 15: Cart Item Subtotal Is Incorrect

- **Endpoint tested:** GET /api/v1/cart (after add)
- **Request payload:**
  - **Step 1 Method:** POST
  - **Step 1 URL:** http://localhost:8080/api/v1/cart/add
  - **Step 1 Body:** {"product$_id$" :< $valid_product_id >, "quantity" : 3\}$
  - **Step 2 Method:** GET
  - **Step 2 URL:** http://localhost:8080/api/v1/cart
  - **Headers:** {"X-Roll-Number": "123", "X-User-ID": "1"}
- **Expected result:** Item subtotal should be exactly quantity * unit$_price.$
- **Actual result observed:** API returned subtotal 104 when expected subtotal was 360 (3 * 120).

Bug 16: Cart Total Does Not Include All Item Subtotals

- **Endpoint tested:** GET /api/v1/cart (after adding multiple items)
- **Request payload:**
  - **Step 1 Method:** POST
  - **Step 1 URL:** http://localhost:8080/api/v1/cart/add
  - **Step 1 Body:** first product quantity 1

  - **Step 2 Method:** POST
  - **Step 2 URL:** `http://localhost:8080/api/v1/cart/add`
  - **Step 2 Body:** second product quantity `2`
  - **Step 3 Method:** GET
  - **Step 3 URL:** `http://localhost:8080/api/v1/cart`
  - **Headers:** `{"X-Roll-Number": "123", "X-User-ID": "1"}`
- **Expected result:** Cart `total` should equal the sum of all line item subtotals.
- **Actual result observed:** API returned total `120` while calculated total from items was `380`.

Bug 17: Profile Accepts Phone Number Containing Spaces

- **Endpoint tested:** `PUT /api/v1/profile`
- **Request payload:** `{"name":"Alice","phone":"12345 6789"}`
- **Expected result:** 400 Bad Request (phone must be exactly 10 digits, no spaces)
- **Actual result observed:** 200 OK
- **Failure detail:** Whitespace-containing phone number is accepted.

Bug 18: Address Creation Succeeds Even When Pincode Is Missing

- **Endpoint tested:** `POST /api/v1/addresses`
- **Request payload:** `{"label":"HOME","street":"123 Main St","city":"Metropolis"}`
- **Expected result:** 400 Bad Request (required field missing)
- **Actual result observed:** 200 OK
- **Failure detail:** Missing `pincode` does not trigger validation failure.

Bug 19: Address City Lower Bound (Length 2) Incorrectly Rejected

- **Endpoint tested:** `POST /api/v1/addresses`
- **Request payload:** `{"label":"HOME","street":"123 Main St","city":"AB","pincode":"123456"}`
- **Expected result:** 200 OK (inclusive minimum boundary)
- **Actual result observed:** 400 Bad Request
- **Failure detail:** Valid boundary value is rejected.

Bug 20: Address City Upper Bound (Length 50) Incorrectly Rejected

- **Endpoint tested:** `POST /api/v1/addresses`
- **Request payload:** city length 50 with valid label/street/pincode
- **Expected result:** 200 OK (inclusive maximum boundary)
- **Actual result observed:** 400 Bad Request
- **Failure detail:** Valid upper boundary value is rejected.

Bug 21: Cart Add Without `product_id` Returns Wrong Status

- **Endpoint tested:** `POST /api/v1/cart/add`
- **Request payload:** `{"quantity":1}`
- **Expected result:** 400 Bad Request (missing mandatory field)
- **Actual result observed:** 404 Not Found
- **Failure detail:** API reports resource-not-found instead of payload validation error.

Bug 22: Cart Add Without `quantity` Is Accepted

- **Endpoint tested:** `POST /api/v1/cart/add`
- **Request payload:** `{"product_id":1}`
- **Expected result:** 400 Bad Request (missing mandatory field)
- **Actual result observed:** 200 OK
- **Failure detail:** Missing quantity is accepted.

Bug 23: Cart Update Without `product_id` Is Accepted

- **Endpoint tested:** `POST /api/v1/cart/update`
- **Request payload:** `{"quantity":2}`
- **Expected result:** 400 Bad Request (missing mandatory field)
- **Actual result observed:** 200 OK
- **Failure detail:** Update is processed without target product id.

Bug 24: Cart Remove Without `product_id` Returns Wrong Status

- **Endpoint tested:** `POST /api/v1/cart/remove`
- **Request payload:** `{}`
- **Expected result:** 400 Bad Request (missing mandatory field)
- **Actual result observed:** 404 Not Found
- **Failure detail:** API reports not-found instead of request validation failure.

Bug 25: CARD Checkout Produces Incorrect Payment Status

- **Endpoints tested:** `POST /api/v1/checkout`, `GET /api/v1/orders`
- **Scenario:** Checkout with `{"payment_method":"CARD"}` succeeds.
- **Expected result:** Created order should have `payment_status = "PAID"`.
- **Actual result observed:** Latest order has `payment_status = "PENDING"`.

Bug 26: Roll Number Header Accepts Leading + Sign

- **Endpoint tested:** `GET /api/v1/admin/users`
- **Request header:** `X-Roll-Number: +123`
- **Expected result:** 400 Bad Request (strict numeric header format)
- **Actual result observed:** 200 OK
- **Failure detail:** Non-strict numeric format is accepted.